Throughput, 200 PU200 ttbar events (RelVal files 3+4), 4 threads, arm (cmssw-el9-native-1)

quantity	0	1	2
mode	cnn	payload	emu
events	200	200	200
steady s/event (begin-processing spacing, events > 20)	9.82	2.46	8.23
steady events/s	0.1018	0.4059	0.1215
first 20 events [s]	190.1	49.6	160.5
job wall [s]	2023.4	539.2	1705.5
job CPU user+sys [s]	7670.7	1684.6	6407.9
CPU/event (job avg) [s]	38.35	8.42	32.04
peak RSS [GB]	6.94	6.95	7.0
FastTimer event loop real [s]	7918.4	2019.2	6632.6
ratio to cnn_nano_v2 (cnn)	1.0	0.251	0.838

Top-12 modules per mode (FastTimerService, all events)

mode	module	type	real/event [ms]	cpu/event [ms]	events
cnn	spixSmartPixelsRecHits	SmartPixelsRecHitProducer	30481.7	30399.5	200
emu	spixSmartPixelsRecHits	SmartPixelsRecHitProducer	24278.8	24211.2	200
payload	other	other	3556.8	2548.5	200
cnn	source	PoolSource	3321.7	2762.1	200
emu	source	PoolSource	3125.3	2558.6	200
emu	NANOAODSIMoutput	NanoAODOutputModule	2804.7	2760.7	200
cnn	NANOAODSIMoutput	NanoAODOutputModule	2751.8	2709.6	200
payload	NANOAODSIMoutput	NanoAODOutputModule	2389.8	2323.2	200
payload	source	PoolSource	2008.9	1505.2	200
cnn	other	other	1036.6	574.3	200
emu	other	other	993.3	597.7	200
payload	l1tTTTracksFromExtendedTrackletEmulation	L1FPGATrackProducer	575.0	571.6	200
emu	idle	idle	549.8	352.4	200
emu	l1tTTTracksFromExtendedTrackletEmulation	L1FPGATrackProducer	543.4	541.6	200
cnn	l1tTTTracksFromExtendedTrackletEmulation	L1FPGATrackProducer	538.0	535.6	200
cnn	idle	idle	472.6	396.8	200
payload	idle	idle	452.7	789.7	200
cnn	cleanup	cleanup	348.3	346.8	200
payload	cleanup	cleanup	275.7	273.2	200
emu	cleanup	cleanup	260.4	259.3	200
payload	spixSmartPixelsRecHits	SmartPixelsRecHitProducer	217.5	214.6	200
payload	spixPixelClusters	SiPixelClusterProducer	110.2	106.3	200
cnn	spixPixelClusters	SiPixelClusterProducer	105.5	103.4	200
payload	l1tTTTracksFromTrackletEmulation	L1FPGATrackProducer	103.0	102.1	200
emu	spixPixelClusters	SiPixelClusterProducer	99.5	96.8	200
emu	l1tTTTracksFromTrackletEmulation	L1FPGATrackProducer	97.6	97.2	200
cnn	l1tSmartPixelsTrackProducerExtendedWdigiRefitAAAA	L1SmartPixelsTrackProducer	94.7	94.1	200
cnn	l1tTTTracksFromTrackletEmulation	L1FPGATrackProducer	94.4	93.9	200
emu	l1tSmartPixelsTrackProducerExtendedWdigiRefitAAAA	L1SmartPixelsTrackProducer	75.9	75.7	200
payload	spixPixelRecHits	SiPixelRecHitConverter	68.9	67.9	200
cnn	spixPixelRecHits	SiPixelRecHitConverter	68.2	67.3	200
payload	l1tSmartPixelsTrackProducerExtendedWdigiRefitAAAA	L1SmartPixelsTrackProducer	68.2	67.6	200
emu	spixPixelRecHits	SiPixelRecHitConverter	66.0	64.9	200
payload	l1tSmartPixelsTrackProducerWdigiRefitAAAA	L1SmartPixelsTrackProducer	62.2	61.8	200
cnn	l1tPh3SmartPixelsClusterTable	L1SmartPixelsClusterTableProducer	61.3	60.8	200
emu	l1tPh3SmartPixelsClusterTable	L1SmartPixelsClusterTableProducer	57.8	57.1	200

SmartPixels modules (spix*) per mode

mode	module	type	real/event [ms]	cpu/event [ms]	events
cnn	spixPixelClusters	SiPixelClusterProducer	105.5	103.4	200
cnn	spixPixelRecHits	SiPixelRecHitConverter	68.2	67.3	200
cnn	spixSmartPixelsRecHits	SmartPixelsRecHitProducer	30481.7	30399.5	200
payload	spixPixelClusters	SiPixelClusterProducer	110.2	106.3	200
payload	spixPixelRecHits	SiPixelRecHitConverter	68.9	67.9	200
payload	spixSmartPixelsRecHits	SmartPixelsRecHitProducer	217.5	214.6	200
emu	spixPixelClusters	SiPixelClusterProducer	99.5	96.8	200
emu	spixPixelRecHits	SiPixelRecHitConverter	66.0	64.9	200
emu	spixSmartPixelsRecHits	SmartPixelsRecHitProducer	24278.8	24211.2	200

SmartPixelsRecHitProducer emu steps (wall time summed over the 4 streams)

mode	step	total [s]	per event [ms]
emu	helix (class-T test)	42.54	212.72
emu	CNN (ONNX)	2926.47	14632.34
emu	donor kNN	1707.12	8535.59
emu	dump kNN	75.5	377.52
emu	draw+bookkeeping	36.36	181.82
emu	one-time setup (ONNX sessions, library load, KD-tree builds)	5.04	nan